

Universität Siegen
Naturwissenschaftlich-Technische Fakultät
Department Elektrotechnik und Informatik
Lehrstuhl für Regelungs- und Steuerungstechnik

Studienarbeit

**Planning, building, and programming a hardware adapter
between Arduino and Lego components.**

by
Milton Sebaastin Clement Sepasthiyammal

Matriculation number
1702059

Examiner
Prof. Dr.-Ing. habil. Michael Gerke

2nd Examiner
Dipl.-Ing. Peter Sahm

Day of delivery

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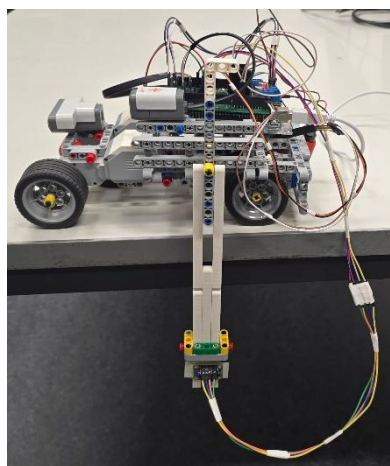
ABSTRACT

This project bridges the gap between LEGO robotics and Arduino by designing and developing a custom adapter shield that allows direct interfacing of LEGO motors and sensors with an Arduino Mega 2560. The primary motivation of this project came from the idea to combine LEGO's modular components with Arduino's flexibility which enables the creation of a mobile robot that is easy to assemble, customize, and control. The custom shield manages power distribution and supports various communication protocols including I2C. The core components include a dual H-bridge L298N motor driver for motor control, a BNO055 orientation sensor for tracking pitch angle, and LEGO touch, light, and ultrasonic sensors for interaction with the environment. The robot responds to touch inputs to reverse direction, detects obstacles using the ultrasonic sensor, and adjusts behaviour in real time. Overall, this project aims to establish a possibility of building a mobile robot by uniting two different ecosystems, making it a valuable foundation for further experimentation in embedded robotics.

1. INTRODUCTION:

Recent methods of prototyping have made the use of LEGO elements together with open-source microcontrollers. While the LEGO components are mechanically robust and ideal for prototyping, their closed hardware ecosystem can be limiting when attempting to incorporate external electronics or control methods.

We have decided to use Arduino, an open-source electronics platform that has become a cornerstone in the maker and educational communities. Known for its flexible hardware, extensive library support, and compatibility with a wide range of sensors and actuators, Arduino offers unparalleled potential for custom development and experimentation. However, Arduino lacks direct mechanical and electrical compatibility with LEGO's proprietary sensors and motors, making integration between these two systems non-trivial.



Pic 1: Image of the Robot

The following report details the complete design and implementation process, including hardware integration, software logic, sensor calibration, circuit design, and experimental validation.

1.1. PROJECT OVERVIEW

The goal of this project is to plan and build a hardware adapter between LEGO components and Arduino to develop a mobile robot. Although LEGO and Arduino platforms are vastly used in prototyping and education, they have different functions and use different hardware standards. While the Arduino boards offer an open and flexible approach to embedded systems development, the LEGO motors and sensors are designed for plug-and-play use within their MINDSTORM controllers. The problem we face here is that these ecosystems are not directly compatible. The development of a hardware adapter between the Arduino MEGA 2560 and various LEGO components, including motors, encoders, and sensors is addressed and taken on in this project. By developing a dedicated adapter board and associated firmware, the two ecosystems which are originally incompatible with each other works seamlessly and provide precise and reliable control of LEGO motor and data acquisition from LEGO sensors using a modern microcontroller.

The foundation of this adapter is a custom designed shield that directly connects with the Arduino Mega 2560 microcontroller. With this custom shield, the LEGO components can be connected directly to the Arduino both electrically and physically. The adapter's design takes connector compatibility, signal voltage levels, and power supply requirements. It includes pin headers for multiple LEGO sensor ports and motor connections. Special attention is paid to I2C communication protocols to facilitate connection between LEGO sensors and Arduino which includes the ultrasonic distance sensor which is operated over the I2C bus. The shield includes routing for I2C lines, and power distribution to support stable and safe operation of all components.

The mobile robot developed using the adapter is has equipped with four LEGO EV3/NXT motors and a combination of sensors to interact with its environment. The sensors consist of a LEGO touch sensor to trigger direction changes, a LEGO ultrasonic sensor for obstacle detection, a LEGO light sensor for surface detection, and a BNO055 gyroscope sensor for measuring and controlling oscillation. The BNO055 sensor is used to measure the oscillations of a loaded arm that is suspended vertically at the robot's side. These measurements are the inputs for a PID controller implemented in Arduino code, which adjusts the motor speed dynamically to stabilize the load while the robot is in motion.

This project also covers the development of an Arduino C code to process the inputs from the sensor and implements speed control based on the sensor feedback. The motor drivers are controlled using PWM signals to regulate speed, and directional control is achieved through digital pin toggling. This code allows all sensor inputs to communicate with one another flawlessly.

Overall, this project demonstrates a way to combine LEGO's straightforwardness with Arduino's flexibility. The entire development cycle from design and fabrication of the PCB to software implementation and testing is discussed in this project. The result of this work is a hardware adapter that can integrate LEGO components in a microcontroller environment and provides the option for future extensions.

1.2. OBJECTIVES AND SCOPE

This interdisciplinary approach combines elements of electronics and embedded programming to deliver a practical solution for educational and prototyping applications. The primary objective of this project is to develop a platform that combines the mechanical modularity of LEGO components with the flexible programming of the Arduino hardware. This project provides direct connectivity of the LEGO motors and sensors with an Arduino Mega 2560 microcontroller by building a custom adapter circuit which is equivalent to an Arduino shield. This platform opens possibilities for advanced control and integration that are typically limited within the standalone LEGO environment.

Specific Objectives:

1. Design a Hardware adapter board:

To develop a printed circuit board (PCB) that can directly be mounted on the Arduino Mega microcontroller. It also provides dedicated pin headers for LEGO sensors and motors and a slot for the dual H – bridge motor driver. The board can provide mechanical connectors compatible with LEGO plugs and safely deliver regulated power to all components.

2. Integration of LEGO sensors

To ensure the connections of LEGO sensors such as touch, light, and ultrasonic with the Arduino which allows seamless communication using GPIO for analog/digital sensors and I2C communication for intelligent sensors like the ultrasonic distance sensor.

3. Implementation of motion control via L298N:

To incorporate the L298N motor driver into the adapter to control the LEGO motors in a differential drive configuration. The control logic supports speed modulation using Pulse Width Modulation [PWM].

4. Load stabilization using sensor feedback and PID control:

To utilize the BNO055 gyroscope sensor to measure the oscillation of a loaded vertical arm. Implement a PID (Proportional-Integral-Derivative) controller to dynamically adjust motor speed using the sensor feedback to compensate the excessive swinging thereby improving stability during movement.

5. Development of an Arduino based control program:

To write an Arduino C code to initialize and read sensors, control the motors, apply PID control, and handle sensor functions like obstacle detection or manual reversal using the touch sensor.

2. HARDWARE COMPONENTS:

2.1. ARDUINO MEGA 2560:

The Arduino Mega 2560 serves as the central processing unit of the robotic system developed in this project. It is an open-source microcontroller board based on the ATmega2560 chip and is widely favoured for applications that require many input/output pins, extensive peripheral connectivity, and adequate program memory. Its digital and analog inputs with dedicated SDA/SCL lines for I2C communication make it particularly well-suited for multi-sensor robotics applications. External (non-USB) power can come from either an AC-to-DC adapter or a battery. The adapter can be connected by plugging a 2.1mm centre-positive plug into the board's power jack. The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts, and the board may become unstable. If using more than 12V, the voltage regulator may overheat and damage the board. Further specifications of the microcontroller can be found in table 1.

The selection of the Arduino Mega over other Arduino variants, such as the Uno or Nano, was driven by the complexity and scale of the project. Given that multiple LEGO sensors and actuators need to be connected simultaneously, including components using I2C communication, digital GPIOs, and PWM control, the Mega's expanded I/O capacity ensures that all peripherals can be interfaced without the need for external multiplexers or GPIO expanders.



Pic 2: Arduino MEGA 2560

Source: Arduino Store

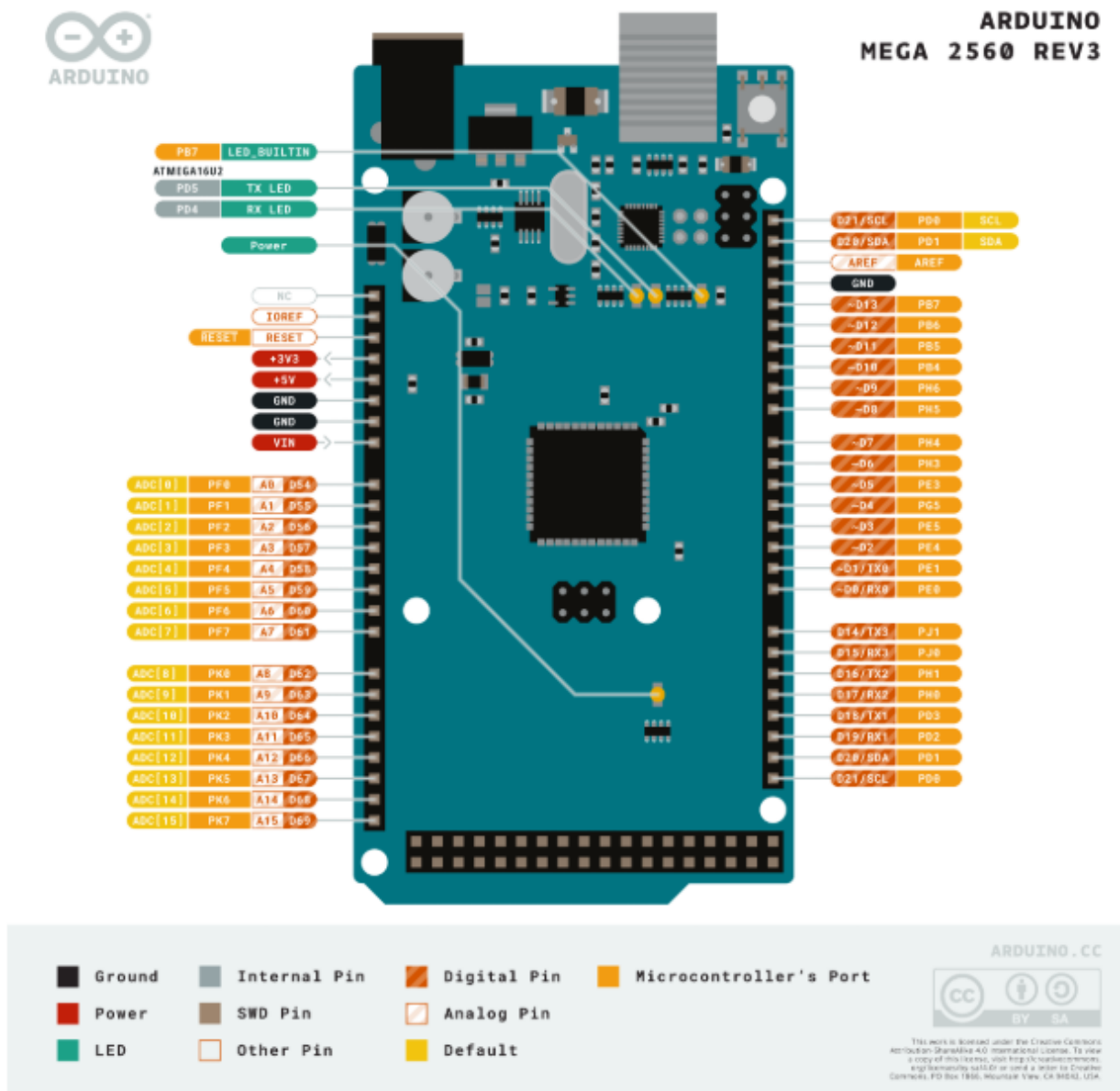
Functionally, the Arduino Mega interfaces directly with the custom adapter shield that was designed as part of this project. This shield is mounted onto the Mega using standard male headers and routes signals from the LEGO-compatible connectors to the appropriate microcontroller pins. Power is supplied to the Arduino via USB during development, and an external DC jack is used during robot operation. The board also regulates power to 5V and 3.3V outputs, which are utilized by various sensors and logic-level converters on the shield.

Parameter	Specification
Operating Voltage	5 V
Recommended Input Voltage	7 V – 12 V (via barrel jack or VIN pin)
Input Voltage Limits	6 V (minimum) to 20 V (maximum)
DC Current per I/O Pin	20 mA (absolute max: 40 mA — not recommended)
Total DC Current for I/O Pins	200 mA (recommended limit for all I/O combined)
DC Current for 3.3 V Pin	50 mA (max, shared with onboard regulator)
Flash Memory	256 KB (of which 8 KB is used by the bootloader)
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz
Power Consumption	~70–200 mA (depending on peripherals and power source)

Table 1: Arduino MEGA 2560 Specifications

Regarding software development, the Arduino Mega is programmed using the Arduino IDE and standard Arduino C libraries. All control logic, including sensor reading, PID computation, motor actuation, and serial communication, is implemented on this board. It is built with support for I2C, PWM, and analog reading, simplifying code integration with LEGO sensors and third-party modules like the BNO055 gyroscopic sensor. Furthermore, the large memory capacity of the Arduino ensures that the code can be developed further as more sensors or complex tasks are added.

In conclusion, the Arduino Mega 2560 is a logical choice for this project due to its hardware capabilities and its scalable and reliable platform is suitable for any further developments. Since it is an open-source platform with significant community support, it makes an excellent prototyping platform for mobile robotics.



Pic 3: Arduino MEGA 2560 Pin configuration

Source: Arduino Store

LEGO EV3 and NXT motors are core actuators in the LEGO MINDSTORMS series, designed specifically for educational robotics applications. These motors are compact in design, precise in mechanical gearing, and have a built-in rotary encoder, allowing high-resolution position and speed feedback. Although initially intended to be controlled by LEGO's proprietary programmable bricks, their simple DC motor structure allows them to be interfaced with third-party controllers such as Arduino using suitable driver circuitry.



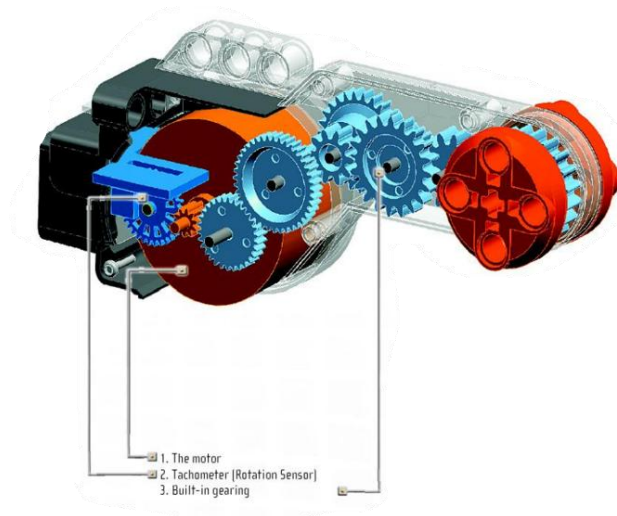
The motors operate on a nominal voltage of 9V DC and draw current suitable for standard H-bridge motor drivers. This project uses four LEGO EV3 motors to implement a differential drive system for the mobile robot. This configuration allows the robot to move forward and reverse.

Direct control of the LEGO motors from the Arduino is difficult due to the current and voltage demands. An **L298N dual H-bridge motor driver** is used as an intermediary to address this. Each EV3 motor is connected to one channel of the L298N, which allows the Arduino to control motor direction via two digital signals and motor speed via a PWM signal.

The motors are connected to the custom adapter shield using T-type female sockets that accept standard LEGO motor cables. These sockets are wired to the output channels of the L298N driver, which in turn receives its control signals from digital I/O pins on the Arduino Mega. Power is supplied to the L298N from an external 9V battery pack to match the operating voltage of the LEGO motors.

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the L298N motor driver and the Arduino Mega controller, the motors are reliable actuators for the mobile robot platform.



Pic 5: LEGO EV3 Motor diagram

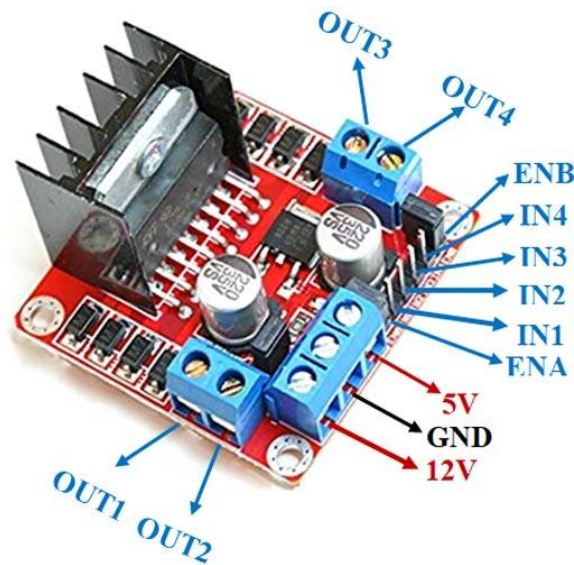
Source: LEGO MINDSTORM User guide

MINDSTORMS EV3 large motor has the following characteristic:

Parameter	Specification
Rated Voltage	9 V DC
Operating Voltage Range	~7.2 V to 10.5 V
Stall Current	~1.2 A
No-Load Current	~150–200 mA
Typical Running Current	~300–600 mA
Maximum Current (peak)	~1.2–1.5 A
Parameter	Encoder Specification
Encoder Type	1° resolution optical encoder
Resolution	360 counts per rotation
Feedback Type	Relative (quadrature encoding)

Table 2: LEGO EV3 Motor Specifications.

2.3. L298N MOTOR DRIVER:



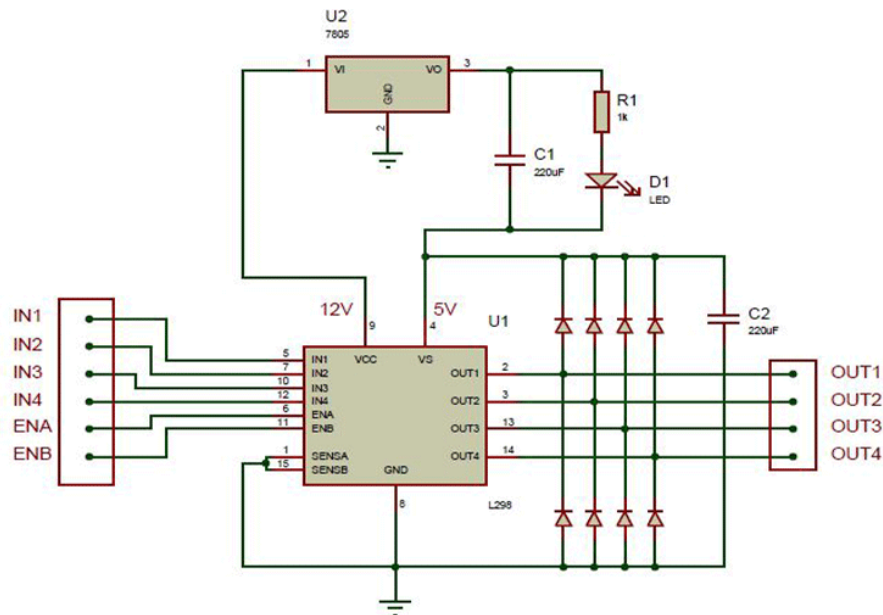
Pic 6: L298N Motor Driver

Source: Components101.com

The L298N is a dual H-bridge motor driver IC that allows for independent control of two DC motors in terms of both speed and direction. It is extensively used in robotics and embedded systems due to its ability to handle high current loads and its compatibility with a broad range of microcontrollers, including Arduino. In this project, the L298N driver serves as a critical intermediary component between the Arduino Mega 2560 and the LEGO EV3 motors, enabling differential drive locomotion for the mobile robot.

This module consists of an L298 motor driver IC and a 78M05 5V regulator. It includes terminal blocks for motor and power connections, heat sinks, and control logic pins. It supports motor voltages up to 46V and output currents up to 2A per channel, which is well-suited for driving LEGO motors, which operate at 9V. The L298N accepts standard TTL (Transistor-Transistor-Logic) level control signals from the Arduino, making integration straightforward.

Each motor connected to the L298N is controlled via two digital input pins (for direction control) and one PWM pin (for speed control). In this setup, one LEGO EV3/NXT motor is assigned to each channel of the L298N. The Arduino Mega generates PWM signals using its `analogWrite()` function and toggles direction pins using `digitalWrite()` commands. For example, setting `IN1=HIGH` and `IN2=LOW` drives Motor A forward, while reversing the logic drives it backward.



Pic 7: L298N Motor Driver Schematic diagram

Source: Components101.com

The motor driver is supplied with an external 9V power source connected to the module's Vcc and GND terminals. The 5V pin on the module is used to power the logic circuitry. In this project, the L298N is powered externally, and the logic signals come directly from the Arduino pins via the custom adapter board. The module also features an onboard voltage regulator and protection diodes for added stability and safety.

Thermal management is important for the L298N during prolonged operation, especially when operating near its current limits. Heat sinks attached to the IC help dissipate heat, and in more demanding environments, additional cooling (such as fans or heatsinks) can be considered. However, in this application with LEGO motors operating below the peak current limit thermal issues have not been encountered.

The L298N's ability to support two motors simultaneously and independently makes it ideal for differential drive systems, where each wheel may be driven separately for turning and manoeuvring. By using separate control lines for the left and right motors, the robot can perform complex movements such as in-place rotation, pivoting, and curved trajectories.

In conclusion, the L298N motor driver provides an efficient solution for controlling LEGO motors through an Arduino. Its electrical and thermal specifications, ease of use, and compatibility with Arduino's PWM outputs make it a key component in this project's actuator control system.

Parameter	Specification
Driver IC	L298N
Motor Channels	2 (dual H-Bridge, for 2 DC motors or 1 stepper motor)
Logic Voltage Range	5 V (can be supplied externally or regulated from onboard 5 V regulator)
Motor Supply Voltage (Vs)	5 V – 35 V (absolute max: 46 V)
Motor Control Logic Voltage	4.5 V – 7 V (recommended: 5 V)
Logic Current Consumption	~36 mA (for control logic)
Max Continuous Motor Current per Channel	2 A (with heat sink and good cooling)
Peak Motor Current (per channel)	3 A (short bursts only)
Output Power (Max)	~25 W total (with heat sink and proper cooling)
Built-in Voltage Regulator	5 V (used to power control logic if Vs > 7 V)

Table 3: L298N Motor Driver Specifications

2.4. LEGO DISTANCE SENSOR:

The LEGO Ultrasonic Sensor is a digital distance sensor used to measure the proximity of objects using sound waves. It functions similarly to commercial ultrasonic sensors but is designed with an exclusive communication interface compatible with LEGO MINDSTORMS systems. In this project, the ultrasonic sensor is critical in enabling the robot to detect and avoid obstacles from point A to point B.

Unlike traditional ultrasonic modules that use trigger and echo pins, the LEGO ultrasonic sensor communicates over the I2C protocol. The sensor is connected to the Arduino Mega via the custom adapter shield, using digital pins 20 (SDA) and 21 (SCL), the dedicated I2C lines on the Mega.

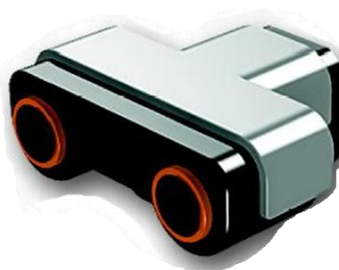
The ultrasonic sensor enables making distance measurement up to 250 cm (100 Inches) in either centimetres or inches. It also enables a listening mode where the ultrasonic sensor measures whether another ultrasonic sensor is active within its surroundings. This sensor is implemented as a digital sensor, meaning the sensor includes a small 8-bit microcontroller that communicates using I2C communication on pins 5 and 6. The ultrasonic sensor sends back a new distance measurement whenever a new distance is detected. If the distance is 250 cm (100 Inches), the duration between new sensor values will approximately be every 15 mS. With a smaller distance, the ultrasonic sensor will be able to return measurements faster. The sensor continuously measures the distance to the nearest object and returns the value with a 1 cm resolution and with an accuracy of ± 3 cm. The red light is continuously turned on around the transducers in the sensor. The ultrasonic sensor measures the distance to an object by sending an ultrasonic sound signal, 12 sound bursts at 40 KHz. The receive transducer is most sensitive at 40 kHz and has a beam angle of approximately 90 degrees.

The ultrasonic sensor is powered using an external 9V source, consistent with LEGO specifications. A voltage regulator and level shifter on the adapter board ensure that the I2C signals from the Arduino (operating at 5V logic) are safely translated for the sensor's internal circuitry. Proper pull-up resistors on the SDA and SCL lines are included in the shield design to maintain bus stability.

Functionally, the sensor emits a high-frequency sound pulse and listens for the reflected echo. It calculates the distance to the nearest object based on the time delay between emission and reception. The Arduino communicates with the sensor by sending read commands to its I2C address and requesting a byte of data corresponding to the measured distance in centimetres.

The robot continuously measures the distance using the ultrasonic sensor and if an obstacle is detected within a set predefined threshold, the robot immediately stops to prevent collisions. This logic allows the robot to operate safely in dynamic environments or during demonstrations.

There was a problem with the I2C communication when paired with the BNO055 gyroscope sensor since the Ultrasonic sensor does not strictly follow the standard I2C protocol. Even the software I2C does not work for the Ultrasonic sensor. But it can be used individually without the combination with the BNO055 gyroscope.



Pic 8: LEGO NXT Ultrasonic Sensor

Source: LEGO MINDSTORM User guide

Parameter	Specification
Operating Voltage	4.3 V DC (from NXT sensor port)
Current Consumption	~20 mA (typical)
Maximum Current Draw	< 30 mA
Communication Interface	Digital – I ² C protocol
I ² C Address	0x02 (default, 7-bit format)
Connector Type	6-pin RJ12 (standard LEGO sensor connector)
Signal Voltage Levels	4.3 V logic (compatible with NXT brick I ² C)
Measurement Range	0 – 255 cm (effective: ~5 cm to 250 cm)
Measurement Resolution	1 cm
Measurement Accuracy	±3 cm (typical)
Update Rate	~20 Hz (one reading every ~50 ms)
Sound Frequency	~40 kHz

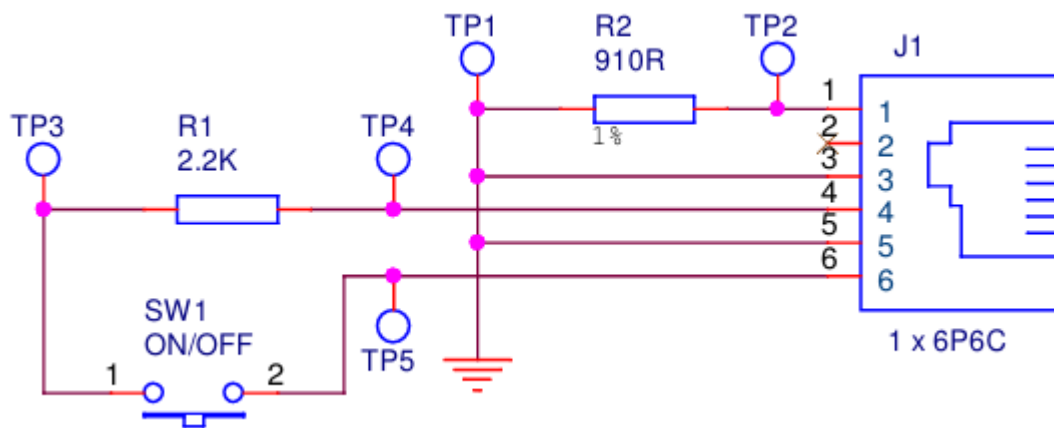
Table 4: LEGO NXT Ultrasonic sensor Specifications

In summary, the LEGO ultrasonic sensor is an effective distance-measuring device that integrates seamlessly with the Arduino via I2C communication. This sensor improves the robot's autonomy and safety, making it an essential component in this project.

2.5. LEGO TOUCH SENSOR

The touch sensor is a very simple sensor which basically enables building a robot which can feel. The sensor consists of a basic mechanical switch which can either be pressed or released. Given the simplicity of the sensor, this sensor is also implemented as a pure digital sensor. The touch sensor plays a critical role in this project by enabling direction reversal functionality, which allows the robot to change its travel path in response to external interactions.

Structurally, the sensor includes a spring-loaded push button encased in a durable housing, making it suitable for frequent or mechanical contact. Electrically, it behaves as a momentary switch that completes a circuit when pressed. In the context of this project, the sensor's output is connected to a digital pin of the Arduino Mega through the custom adapter board.



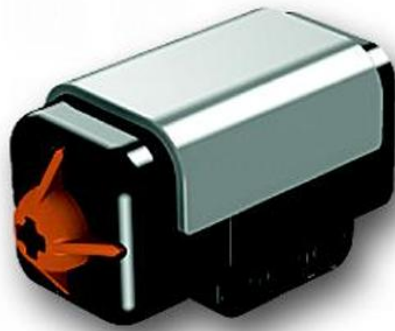
Pic 9: LEGO EV3 Touch sensor Schematic diagram

Source: LEGO MINDSTORM Hardware Developer Kit

The adapter routes the sensor's signal line and provides a ground reference via a standard LEGO-compatible socket. The sensor operates at 5V logic levels and does not require any special communication protocols or libraries. It is read in the Arduino code using `digitalRead()`, returning HIGH when pressed and LOW when released.

The touch sensor is programmed to toggle the robot's motion direction each time it is pressed. This feature is to provide a manual override or edge detection mechanism, especially useful in constrained environments. When pressed, the robot reverses its motor control logic, changing its direction of travel without requiring positional feedback from the ultrasonic sensor.

The touch sensor is also helpful in handling collision scenarios. For example, if the robot contacts a wall or another object that presses the sensor, it can be programmed to reverse or turn away. This reactive capability adds adaptability to the system, primarily when operating in semi-structured environments.



Pic 10: LEGO Touch Sensor

Source: LEGO MINDSTORM User guide

Though only two touch sensor is used in the current setup, multiple sensors can be integrated to provide feedback on different sides of the robot. This would allow for more sophisticated collision handling or multi-point user control interfaces.

The LEGO Touch Sensor provides a simple yet highly effective method for triggering user or environment-driven actions. Its reliability, ease of integration, and physical durability make it a valuable addition to the LEGO-Arduino mobile robot platform.

Parameter	Specification
Operating Voltage	4.3 V DC
Current Consumption	< 0.5 mA (very low, passive switching)
Signal Voltage (HIGH)	~4.3 V (button pressed)
Signal Voltage (LOW)	0 V (button not pressed)
Interface Type	Digital – via analog voltage level detection
Communication Protocol	None (simple analog voltage change read as digital signal)

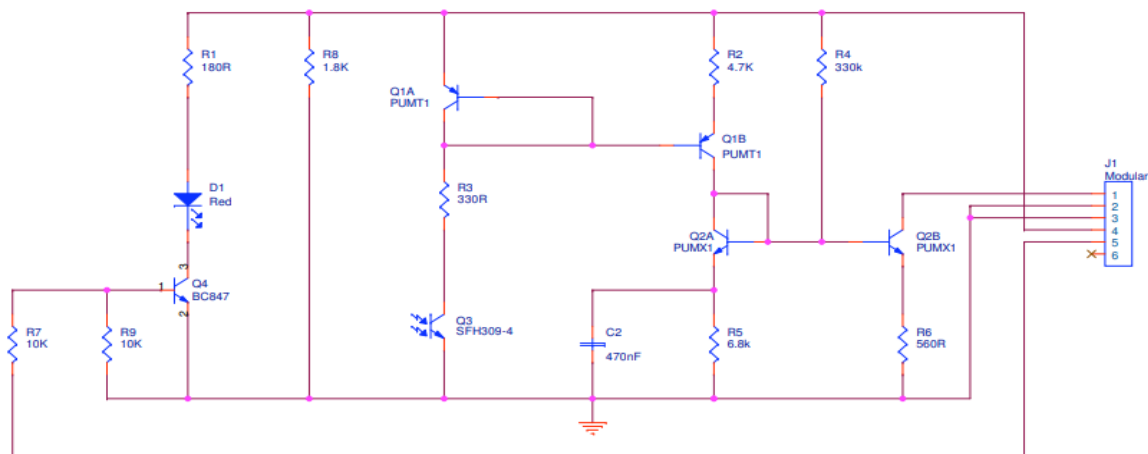
Table 5: LEGO Touch Sensor Specifications

2.6. LEGO LIGHT SENSOR

The Light Sensor is one of the sensors that gives the robot system vision. The Light sensor enables the robot to distinguish between light and dark. It can read light intensity in a room and measure the light intensity reflected from surfaces.

The LEGO Light Sensor is a fundamental sensing component in the MINDSTORMS NXT system, designed to detect the intensity of ambient or reflected light. It enables basic environmental perception by differentiating between surface colours, detecting lines on the ground, or responding to changes in lighting conditions. The light sensor provides valuable input for future navigation enhancements and environmental awareness in this project.

The light sensor emits a beam of light—typically from a red LED—and measures the amount of light reflected using a phototransistor. The sensor is implemented as a digital sensor. In this project, the sensor is connected to the Arduino via a general-purpose input pin capable of reading analog signals. The sensor is powered through the custom adapter shield, which provides 5V power and ground through LEGO-compatible T-type connectors.



Pic 11: LEGO Light Sensor Schematic diagram

Source: LEGO MINDSTORM Hardware Developer Kit

Although the primary focus of the robot is not line-following, the light sensor has been integrated to allow for potential extensions such as surface-type detection (e.g., light tile vs. dark carpet) or environmental triggers based on colour transitions. For example, the robot could use the light sensor to slow down in dimly lit areas to reduce oscillation risk.

From an electrical perspective, the light sensor is straightforward to interface with Arduino. The required connections are ground, 5V supply, and a signal line connected to an analog input on the Arduino Mega. The signal voltage ranges between 0V and 5V, depending on the intensity of the reflected light. The Arduino reads the signal, and threshold values can be used in code to interpret surface changes or trigger behavioural responses.



Pic 12: LEGO Light Sensor

Source: LEGO MINDSTORM User guide

Initial calibration is required during setup for consistent readings as the sensors measurement is affected by external lighting conditions. This involves measuring light levels on known surfaces (e.g., white vs. black) and defining decision thresholds accordingly. These values can then be used accordingly during operation.

While not critical to the robot's core functions—such as obstacle avoidance or load stabilization—the light sensor offers a low-cost way to improve contextual awareness. Its integration demonstrates the flexibility of the adapter board and opens opportunities for adding autonomous features or path-following logic in future versions of the system.

There are three modes for the Light sensor.

Light sensor mode: Within this mode the sensor turns on the red led and measures the reflected light intensity. The optimal reading distance is between 0.5 cm and 1.5 cm LEGO module distance. The sensor automatically cancels out any backlight. This is done by very quickly turning the red led off and measure the backlight which is then subtracted from the actual reflected reading. This is continuously done 1000 times/sec to enable 1000 sample/sec. The output from the sensor within light sensor mode is a value between 0 – 100 where 0 equal no reflected light measured and 100 equals full light reflection.

Ambient sensor mode: Within this mode the sensor can read ambient light within its surroundings. The sensor turns on the blue led in this mode to visually indicate which mode it is in. Given that the sensor uses an algorithmic conversion table the sensor can measure from very dark to very bright sun light. The output from the sensor within ambient sensor mode is a value between 0 – 100 where 0 equal darkness while 100 equals directly pointing towards sunlight.

Colour sensor mode: Within this mode the sensor can determine the colour on the object in front of it. Again, the optimal reading distance is between 0.5 cm – 1.5 cm LEGO module distance. Within this mode the sensor automatically quickly reads the individual reflected light intensity for red, green, blue and the backlight. Using these values the sensor can determine between 7 different colours and whether no object is observable. The 7 different colours are: black, blue, green, yellow, red, white and brown.

Parameter	Specification
Operating Voltage	4.3 V
Current Consumption	~3–6 mA (idle); <20 mA (LED active during measurement)
Power Supply Tolerance	4.2–4.4 V
Signal Output Type	Analog voltage (0–5 V, typically 0–3.3 V depending on light intensity)
Typical Output Range	0.0–3.0 V
Sensor Type	Phototransistor + Red LED
Output Impedance	~10 k Ω (estimated from op-amp buffered output stage)
Sampling Frequency	Up to ~1 kHz (internal ambient subtraction and LED toggling at high frequency)

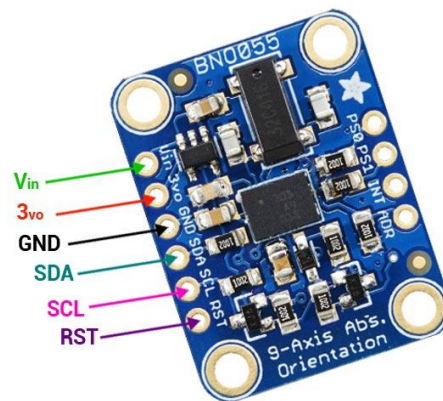
Table 6: LEGO Light Sensor Specifications

2.7. BNO055 GYROSCOPE:

The BNO055 is a highly advanced 9-axis absolute orientation sensor developed by Bosch Sensortec. Unlike traditional inertial measurement units (IMUs) that require sensor fusion algorithms to interpret raw data, the BNO055 features an integrated microcontroller that performs onboard sensor fusion. It combines data from a 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer to output absolute orientation in the form of Euler angles. This functionality greatly simplifies development and improves accuracy in motion-sensitive applications. This chip can also be found in a prebuilt module, made by Adafruit industries. As this module comes with a generic I2C interface, communication with different microcontrollers and FPGA becomes easier.

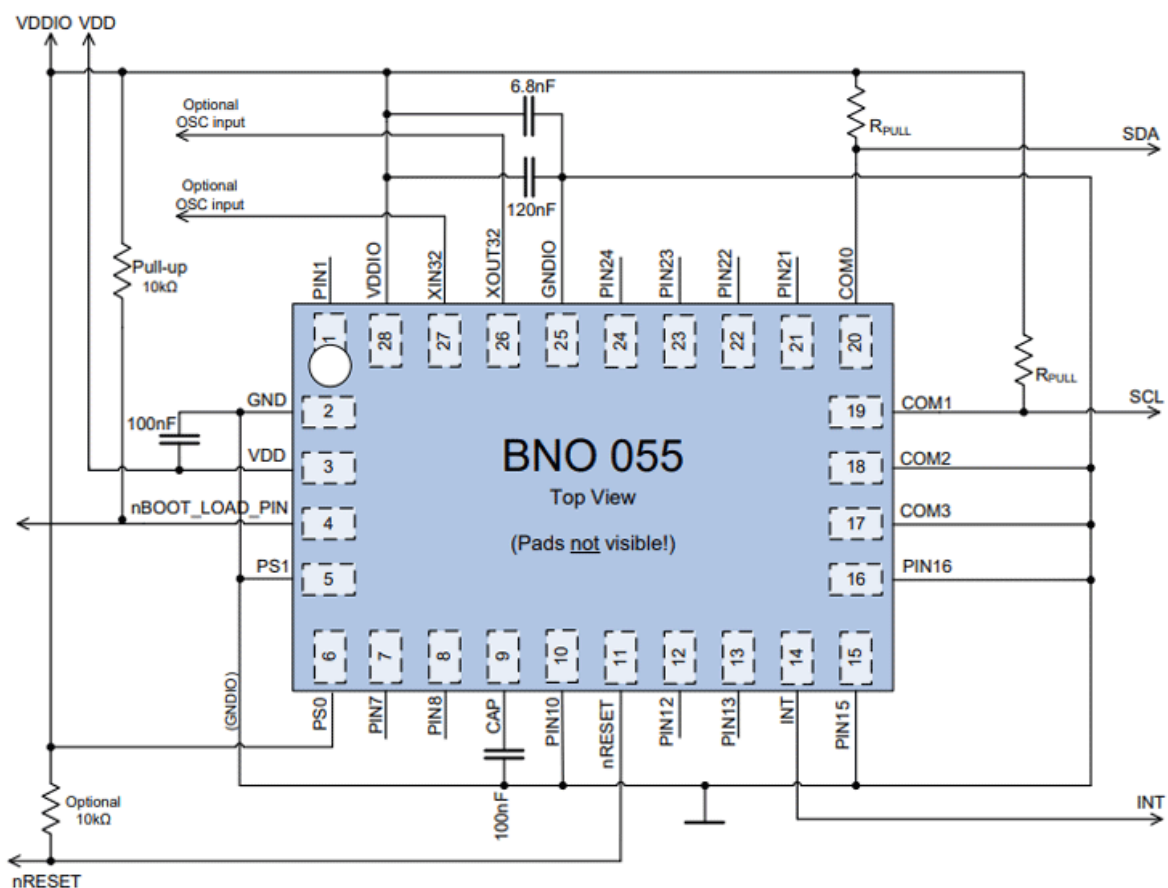
In this project, the BNO055 plays the main role in measuring and managing load oscillation. The robot includes a vertical arm that carries a suspended load, which is prone to oscillate when the robot moves. By continuously measuring the oscillation of the arm using the BNO055, the system dynamically adjusts motor speed using a PID controller to stabilize the load during motion. This forms a closed-loop feedback system ensuring safety and stability of the robot carrying fragile components.

The BNO055 communicates with the Arduino Mega via the I2C protocol. The I2C bus was intended to be shared with the LEGO ultrasonic sensor, but both devices did not operate successfully on the same bus due to their unique I2C dependencies. Power is supplied through the 5V pin on the Arduino, and both PS0 and PS1 pins on the BNO055 are grounded to force the device into I2C mode.



Pic 13: BNO055 Sensor

Source: Components101.com



Pic 14: BNO055 Sensor Schematic diagram

Source: Components101.com

The sensor is included in the Arduino code using the Adafruit_BNO055 library, which provides functions to configure output format, read Euler angles, and monitor calibration status. The pitch angle is used as the feedback variable in the PID control loop. This feedback is compared to a target pitch angle, and the resulting error is used to modulate motor speed to dampen the oscillation.

One key advantage of the BNO055 is its high immunity to drift and noise. The onboard algorithm includes filtering and calibration routines that provide stable readings even during extended operation. The measured data is stable and reliable with minimal latency which makes it ideal for real-time control applications. This sensor can operate in multiple modes, such as NDOF (Nine Degrees of Freedom), IMU-only, and compass mode. In this project, the NDOF mode is used to measure the angles.

In summary, the BNO055 sensor makes the mobile robot unique by measuring accurate orientation and real-time stabilization of a suspended load. Its ease of integration, onboard fusion processing, and reliability make it an important part of the robot's sensor group.

Parameter	Specification
Sensor Type	9-DOF IMU (Accelerometer + Gyroscope + Magnetometer + MCU for fusion)
Operating Voltage (VDD)	2.4 V to 3.6 V (typically 3.3 V)
I/O Voltage (VDDIO)	1.8 V to 3.6 V (separate pin, often tied to 3.3 V)
Current Consumption (Normal Mode)	12 mA (typical in fusion mode)
Current Consumption (Suspend Mode)	40 μ A
Logic Level	3.3 V logic (NOT 5V tolerant on any pin!)
Communication Interfaces	- I ² C (up to 400 kHz)- UART- SPI (up to 10 MHz)
Default I²C Address	0x28 (can be switched to 0x29 via COM3 pin)
Output Data Rate	Up to 100 Hz (in fusion mode)

Table 7: BNO055 Sensor Specifications

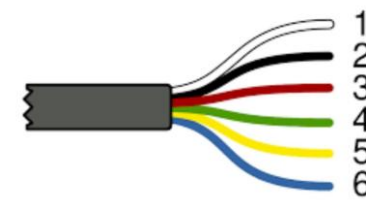
3. CIRCUIT CONNECTIONS:

The following part discusses the design and wiring of the components with the microcontroller

3.1. RJ12 WIRE:

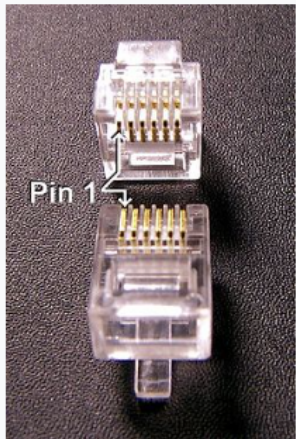
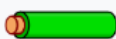
The LEGO sensors and motors use a 6-pin RJ12 modular connector. While the connector pin is similar to the standard RJ12 plug used in telecommunications in terms of its appearance, the internal pin assignment and colour scheme follow proprietary wiring scheme. Clear understanding of this colour mapping and its functions is necessary when designing the adapter that interfaces the Arduino and the LEGO components.

The colour code used in the LEGO cables typically follows the pattern as shown below:



Pic 14: LEGO Cable Colour Code

Source: Wikipedia

Pin	Name	Function	Color	Pin number
1	ANALOG	Analog interface, +9V supply	 White	
2	GND	Ground	 Black	
3	GND	Ground	 Red	
4	IPOWERA	+4.3V supply	 Green	
5	DIGIAI0	I ² C clock (SCL), RS-485 B	 Yellow	
6	DIGIAI1	I ² C data (SDA), RS-485 A	 Blue	

Pic 15: LEGO Wire Pin Configuration

Source: Wikipedia

3.2. MOTOR CONNECTION:

The layout of the Motor connection is a crucial part of the system's wiring design, as it determines the interface between the motors and the Arduino Mega 2560 through the L298N motor driver. Direction and speed control of the motors is achieved by assigning two digital pins and a PWM pin per motor.

The output voltage supply is achieved by connecting the LEGO motors to the output terminals of the L298N as follows: pins OUT1 and OUT2 for Motor A and pins OUT3 and OUT4 for Motor B. The power for the L298N is supplied by connecting the Vcc to an external 9V battery, which can deliver sufficient current and voltage for both motors under load. The GND terminal is shared between the battery pack, Arduino and the motor driver to maintain a common reference voltage. The encoders of the motors are powered by directly connecting them to the digital pins of the Arduino via the shield.

Control signals for Motors A and B are provided through the pins IN1, IN2, and IN3, IN4, respectively. These pins are directly connected to the digital pins of the Arduino via the shield. Each motor receives PWM signals through its dedicated enable pins ENA and ENB for Motors A and B, respectively, which are then connected to the PWM-enabled pins of the Arduino MEGA. The PWM signals allow precise control over motor speed by adjusting the Duty cycle.

Safety and reliability are considered while designing the wiring. The L298N motor driver has built-in flyback diodes to protect the motors against voltage spikes generated due to sudden changes in direction or speed. The high current motor lines are kept physically separate from the sensitive sensor signal lines to minimize noise.

This wiring arrangement enables the Arduino MEGA to perform differential drive control, allowing the robot to move forward, backward, and pivot in place by controlling the speed and direction of the motor individually through Arduino code. This wiring design makes it easy to replace or reconfigure the motors for different robot designs without altering the core control scheme.

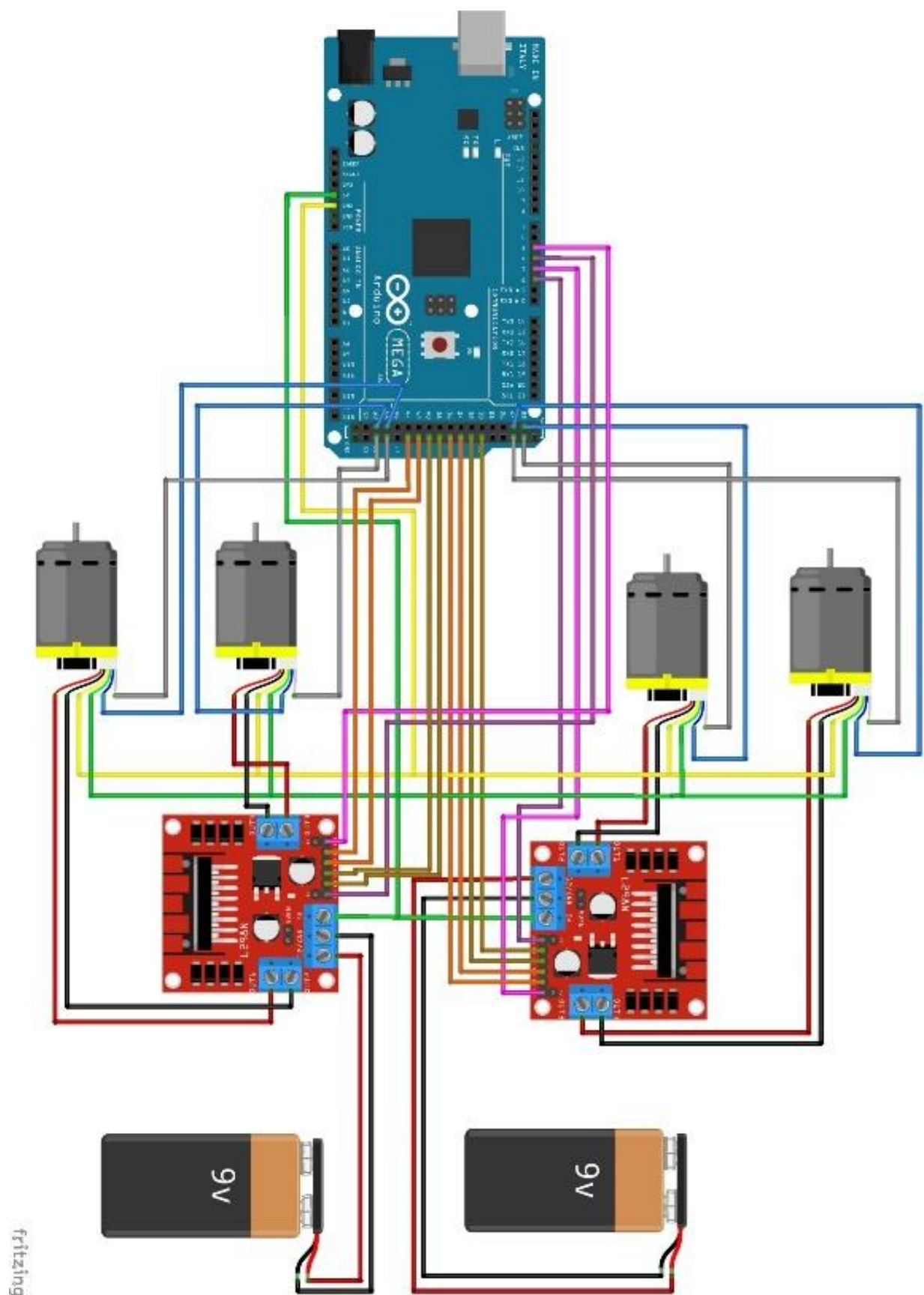
The following table shows the general connections of the LEGO motors to the Arduino MEGA via L298N motor driver. The digital pins connection corresponding to the motors can be varied based on convenience of the user.

Motor A Pin	Pin Colour	Function	Arduino Mega Pin			
1	White	Motor output	OUT 1 [L298N driver]			
2	Black	Motor output	OUT 2 [L298N driver]	L298N Pin	Function	Arduino Mega Pin
3	Red	Ground	GND	ENA	PWM control	PWM pin 2
4	Green	Encoder power	5V	IN1	Control signals	Digital pin
5	Yellow	Encoder	Digital pin	IN2	Control signals	Digital pin
6	Blue	Encoder	Digital pin			

Table 8: Connections for Motor A

Motor B Pin	Pin Colour	Function	Arduino Mega Pin																		
1	White	Motor output	OUT 3 [L298N driver]																		
2	Black	Motor output	OUT 4 [L298N driver]	<table><tr><th>L298N Pin</th><th>Function</th><th>Arduino Mega Pin</th></tr><tr><td>ENB</td><td>PWM control</td><td>PWM pin 3</td></tr><tr><td>IN1</td><td>Control signals</td><td>Digital pin</td></tr><tr><td>IN2</td><td>Control signals</td><td>Digital pin</td></tr><tr><td colspan="3"></td></tr></table>	L298N Pin	Function	Arduino Mega Pin	ENB	PWM control	PWM pin 3	IN1	Control signals	Digital pin	IN2	Control signals	Digital pin					
L298N Pin	Function	Arduino Mega Pin																			
ENB	PWM control	PWM pin 3																			
IN1	Control signals	Digital pin																			
IN2	Control signals	Digital pin																			
3	Red	Ground	GND																		
4	Green	Encoder power	5V																		
5	Yellow	Encoder	Digital pin																		
6	Blue	Encoder	Digital pin																		

Table 9: Connections for Motor B



Pic 16: Motor connections with Arduino through L298N

3.3. SENSOR CONNECTIONS:

a. ULTRASONIC SENSOR:

Sensor Pin	Pin Colour	Function	Arduino Mega Pin
1	White	+9V Supply (external battery)	External 9V IN
2	Black	Ground	GND
3	Red	Ground	GND
4	Green	Power	5V
5	Yellow	I ² C Clock (SCL)	Digital 21 (SCL)
6	Blue	I ² C Data (SDA)	Digital 20 (SDA)

Table 10: Connections for Ultrasonic Sensor

b. TOUCH SENSOR:

Sensor Pin	Pin Colour	Function	Arduino Mega Pin
1	White	Ground	GND
2	Black	Not Connected	N/A
3	Red	Signal Output	Digital 48
4	Green	Not Connected	N/A
5	Yellow	Not Connected	N/A
6	Blue	Not Connected	N/A

Table 11: Connections for Touch Sensor

c. LIGHT SENSOR:

Sensor Pin	Pin Colour	Function	Arduino Mega Pin
1	White	Analog Output (Light Level)	Analog A0
2	Black	Ground	GND
3	Red	Ground	GND
4	Green	+4.3V Supply	5V
5	Yellow	Signal Output	Digital 36
6	Blue	Not Connected	N/A

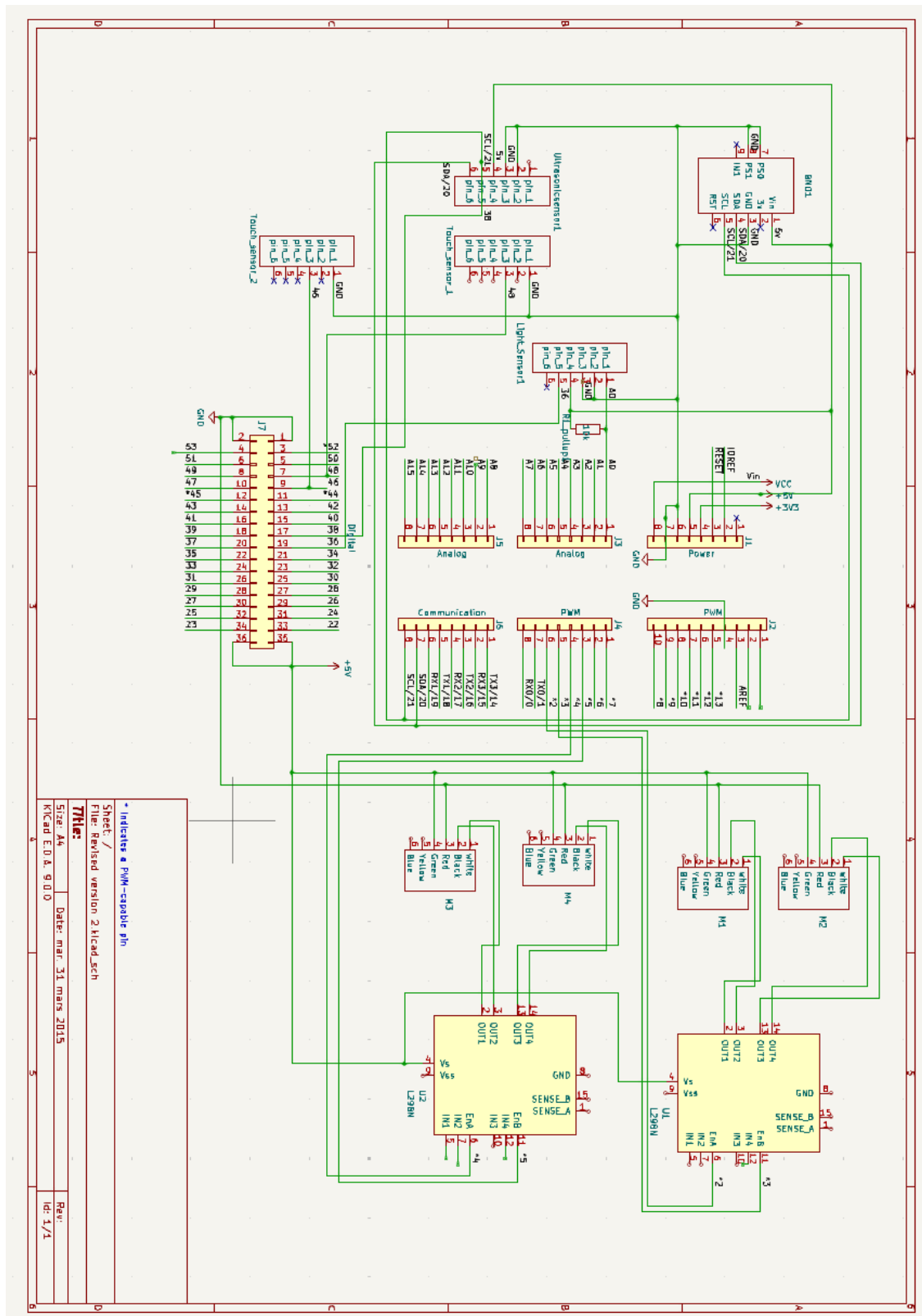
Table 12: Connections for Light Sensor

d. BNO055 GYROSCOPE:

Sensor Pin	Function	Arduino Mega Pin
Vin	+5V Supply	5V
GND	Ground	GND
SDA	I ² C Data	Digital 20 (SDA)
SCL	I ² C Clock	Digital 21 (SCL)
PS0	Mode Select	GND
PS1	Mode Select	GND

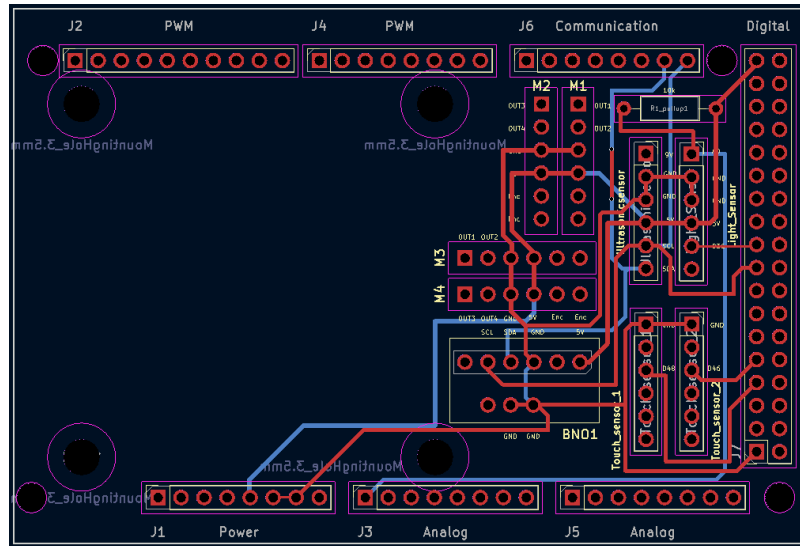
Table 13: Connections for BNO055 Gyroscope

3.4. CIRCUIT DESIGN AND FABRICATION:



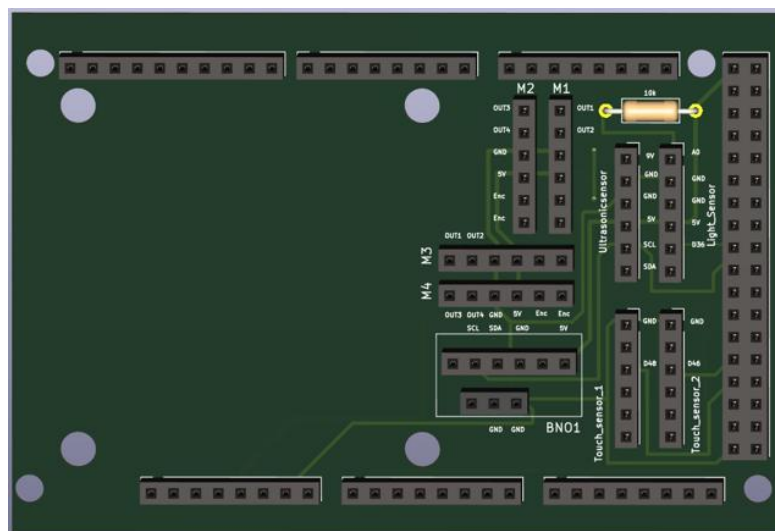
Pic 17: Circuit Design

The above schematic is used for the design and fabrication of the circuit board. The custom adapter PCB is designed using the design application KiCAD. The custom adapter is designed on a 2-layer circuit board containing pads to accommodate the sensors and motor.



Pic 18: Wiring blueprint of the Circuit board

The board is constructed for easy plug and play operation of the sensors, which is made possible by the use of generic header pin connectors.



Pic 19: Image of the fabricated PCB

4. IMPLEMENTATION:

This part of the report discusses the firmware architecture, control logic, and practical considerations for operating the mobile robot. This implementation prioritizes reliable I2C communication between sensors and to determine the control loop for the oscillation damping with motion control and obstacle detection.

4.1. SOFTWARE ENVIRONMENT AND TOOLS

The LEGO mobile robot development relied on an open-source software ecosystem to ensure flexibility, reliability, portability, and ease of use. The programming and debugging part of this project was done on Arduino IDE, which is an integrated development environment for the Arduino microcontrollers.

PRIMARY DEVELOPMENT AND PROGRAMMING FRAMEWORK:

Arduino IDE (Integrated Development Environment) was used to develop the firmware. This environment was selected for the ease code compilation and uploading capabilities, built-in library management and for its community support. The Serial monitor feature was useful for real time debugging and data logging during testing phases. The control code is written in Arduino C language, using the standard Arduino functions such as `digitalWrite()`, `analogWrite()`, and `analogWrite()` for motor and sensor operations. USB communication with the Arduino MEGA was achieved through the board's onboard ATmega16U2 USB-to-Serial converter. The Arduino IDE's serial monitor was extensively used during the development for the sensor output readings and PID control values. I2C scanner script was employed for the I2C address verification to ensure that all devices were correctly connected and responding at their assigned address.

LIBRARIES USED:

- Wire.h – The core Arduino library for I2C communications, used to interface with both LEGO sensors and the BNO055 gyroscope sensor over a shared bus.
- Adafruit_BNO055.h – Provides high level functions for initializing the BNO055 sensor, reading Euler angles and monitoring calibration status.
- Adafruit_Sensor.h – Provides a standardized data access from the sensor.

By combining Arduino IDE's well supported libraries and structured documentation practices, this project achieved a reliable and maintainable software foundation to support the robot's sensor and control requirements.

4.2 BASIC FUNCTIONS AND PROGRAMMING IN ARDUINO C:

The programming follows a modular structure, where each hardware component is formed into its own set of functions. This approach ensures that the motion control, sensor data acquisition, and feedback loop are easy to read, debug, and modify. The following sections deals with the implementation of the basic functions essential for the robot operation.

4.21. MOTOR CONTROL:

The LEGO motors are driven by the Arduino MEGA 2560 via the L298N motor driver. The motor direction is controlled using the digital pins and the speed is regulated using a PWM enabled pin.

Implementation details:

The implementation logic is simple, when the motors spin in the same direction and at the same speed, the robot moves forward, and by reversing the polarity, the motor moves backward. In code, both forward and reverse motion are set into their own functions. These functions set the logic on the L298N's direction pin and applies PWM value to control the motor speed. A dedicated function [stopMotor()] sets all the motor inputs to low and disables the PWM output, which ensures the immediate halting of the robot.

4.2.2. GYROSCOPE DATA ACQUISITION:

The main goal of this project is the mobile robot's stability, which is achieved with accurate motion sensing. The BNO055 sensor provides real time measurement of the robot's pitch, roll, and yaw angles. For this project, only the pitch angle is required, as it directly denotes the oscillation of the loaded vertical arm.

Implementation details:

The BNO055 sensor communicates with the Arduino MEGA via I2C protocol. Once the robot starts up, the Arduino initiates the sensor, sets it to NDOF mode, and then waits for it to reach the desired calibration status. The sensor measures the latest pitch angle and feeds it as an input for the PID controller in a loop. The low noise output of the sensor ensures smooth operation.

4.2.3. ANGULAR DISPLACEMENT CONTROL:

The load stabilization of the robot is based on the PID based angular displacement control. The goal is to keep the vertical load arm as close to upright as possible when the robot moves.

Implementation details:

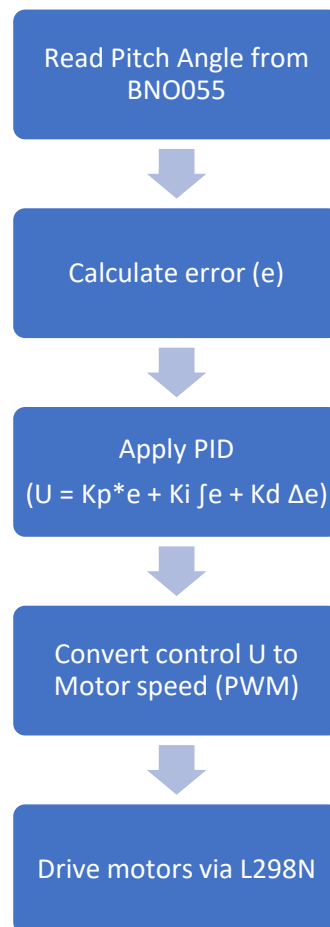
When the measured pitch angle from the BNO055 sensor deviates from the set threshold value, the PID algorithm adjusts the motor speed to reduce the error. Large deviations cause the robot to slow down rapidly, giving time for the vertical arm to settle. The PID gains are tuned so that the robot acts quickly to the error, without overcompensating and causing further oscillation. The P-part of the controller provides immediate correction, and the I-part is included in case there is a consistent tilt due to accumulated error over time. The D-part dampens the sudden changes, thus avoiding overcompensation.

The general PID control law used in this implementation is expressed as:

$$U(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{d e(t)}{dt}$$

The tuning process followed a manual, empirical approach based on system response observation through the Arduino Serial Plotter, where the real-time pitch angle and motor speed (PWM) were monitored.

The following picture shows the relationship between pitch angle and motor speed by the PID controller. The controller continuously monitors the pitch angle, computes the error from the desired setpoint, and adjusts the motor speed proportionally using PID terms to minimize oscillation and maintain stability.



Pic 20: PID flowchart

To summarize, consider the robot is moving, causing the arm to swing, the P-part reacts to the angle, and the D-part reacts to the swing speed, which makes the motor slow down. If the arm has been leaning slightly for several seconds, the I-part adjusts the motor speed further to fix it.

5. TESTING AND RESULTS:

The aim of the testing is to ensure the following functionalities: forward motion, reverse motion, to respond to the obstacles and load stabilization using PID control. The testing procedures was done in three phases discussed below.

5.1. EXPERIMENTAL SETUP:

The experiment was conducted in an indoor environment to avoid external variables like wind and uneven surfaces. The robot configuration is as follows:

- 1) Drive system: LEGO motors connected via L298N motor driver.
- 2) Sensors:
 - a) Adafruit BNO055 orientation sensor - pitch measurement.
 - b) LEGO Ultrasonic sensor - obstacle detection.
 - c) LEGO Touch sensor - manual direction override.
 - d) LEGO Light sensor - future extension.
- 3) Controller: Arduino MEGA 2560 with custom adapter shield.
- 4) Power Supply: 9V battery for motors and Arduino's regulated 5V supply for logic and sensors.

A vertical arm loaded with weight is mounted on the side of the robot to evaluate oscillation control. The robot's actions were configured to move forward until interrupted by an obstacle or the touch sensor is pressed, and to adjust motor speed continuously to minimize oscillations.

5.2. PERFORMANCE OF MOTOR OPERATIONS:

The motor performance was verified by forward and reverse motions, and speed modulation.

- Forward motion: When the function to move the motor forward was called, both the motors moved in the same direction with same speed.
- Reverse motion: When the function to move the motor backward was called, both the motors changed its direction of rotation and achieved constant speed.

Overall, the motor performance met the requirements based on the given inputs.

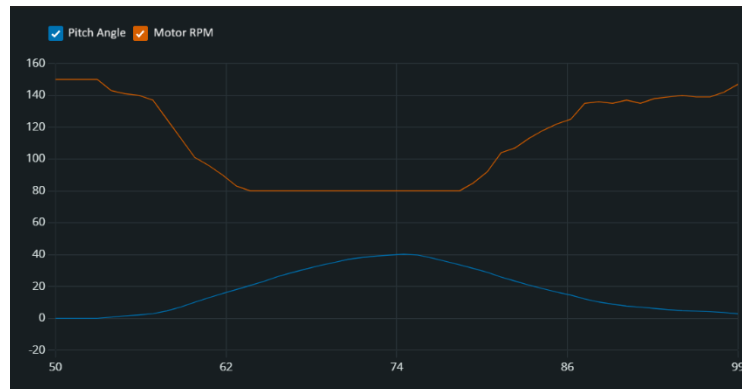
5.3. GYROSCOPE AND SENSOR DATA VALIDATION:

BNO055 Orientation sensor: With the robot stationary, the measured angles remain within $\pm 0.3^\circ$ of the reference, indicating the stable position of the arm. During forward motion, the sensor measured the oscillations of the vertical arm with minimum noise, even during acceleration. The PID controller responded when the oscillations exceeded $\pm 10^\circ$, leading to measurable damping and load stabilization.

LEGO Ultrasonic Sensor: The distance sensor continuously measured the distance between the obstacle and the robot. When the sensor reads value less than 5cm, the motor is forced to stop. I2C communication with the Arduino remained stable despite the shared I2C bus with the orientation sensor.

LEGO Touch Sensor: Produced reliable HIGH/LOW transitions with no false triggering. The motor direction changes every time the sensor is triggered.

LEGO Light Sensor: Detected differences in surface reflectivity and dark and light ambient lightings. While the functionalities are not actively used in the project, the sensor is ready for any future developments.



Pic 21: Graph depicting the relation between Pitch angle and Motor speed

In summary, the sensors performed as expected under the set tolerance. The orientation sensor provided accurate measurements for PID control, the distance sensor ensures quick sensor detection, and the Touch sensor operated without any hitch.

6. CONCLUSION:

This project demonstrates the design, development, and validation of a custom hardware and software platform that integrates LEGO sensors and motors with Arduino microcontroller. The aim of this project is to create a mobile robot capable of moving between two points in a straight line while carrying a suspended weight on the side. This robot also has a capability of dynamically adjusting its speed to control the oscillations using PID controller.

The core of this project is to design a custom shield for the Arduino MEGA to directly interface the LEGO sensors with the said microcontroller. This shield handles the power distribution, I2C communication and routing of signals through RJ12 connectors. This hardware allows multi-sensor operations such as BNO055 orientation sensor, LEGO distance sensor, LEGO touch sensor, and LEGO light sensor. The dual bridge L298N motor driver provided reliable control of the LEGO motors for forward and reverse movements along with speed modulation through PWM control.

Experimentations verified the desired functioning of all systems. The motor control was smooth with synchronized motion and directional changes. The distance sensor detected the obstacles, and the touch sensor reliably triggered directional changes. The BNO055 orientation sensor provided accurate pitch angle measurements after calibration which in turn allowed the PID controller to effectively maintain oscillations. While the robot

performed as expected, certain limitations were observed. The PID parameters required careful tuning.

Future work could include integrating the light sensor to enable line following mechanism or include LEGO sound sensor for voice control. The modular nature of the hardware and software ensures these additional developments be implemented without any major changes.

Overall, this project efficiently bridges the LEGO mechanical systems with Arduino based control suitable for experimentations and innovations.

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